

CHAPTER 6 – SYSTEM TESTING

6.1 Introduction

Testing verifies that a software system operates correctly, reliably, and as specified under real-world conditions. For a safety-critical application like **RakshaSafe**—where citizens trigger SOS alerts, admins monitor live incidents, and users rely on emergency contacts—absolute system dependability is essential.

Because Raksha Safe is built across three cooperating tiers—a React + TypeScript + Vite frontend, a Node.js + Express.js + TypeScript REST API, a MongoDB database, and a Python + Fast API service for AI risk assessment—testing was conducted at all levels: inside the source code, at the API boundary, and end-to-end within a real browser.

The testing process combined structured manual execution with fast automated checks. This included strict TypeScript compilation, oxlint code checks, production builds, programmatic API calls, direct MongoDB document inspection, and complete user/admin journey walkthroughs in Google Chrome.

6.2 Objectives of Testing

- **Functional Verification:** Ensure every Chapter 3 feature operates correctly—user auth, profiles, SOS creation, location handling, nearby facilities, rescue teams, notifications, unsafe-area reports, AI risk assessments, and admin functions.
- **Security & Access Control (RBAC):** Reject unauthenticated requests, restrict admin controls from standard users, and strictly enforce data ownership isolation across all endpoints.
- **Database & Data Integrity:** Inspect actual MongoDB documents to confirm valid record creation, correct data linking, and schema consistency.
- **Input Validation:** Reject invalid or malformed data (phone numbers, emails, passwords, field lengths) with clear error messages without corrupting the database.
- **Usability & Responsiveness:** Validate clear interface labeling, multi-language support (English/Hindi), and fluid UI adaptation from desktop down to mobile screen sizes.
- **Reliability & Resilience:** Confirm incident data persists reliably and ensure third-party service outages (Email, SMS, WhatsApp, Maps, AI) never crash the system or block core operations.
- **Defect Lifecycle Management:** Record, fix, and re-verify every identified defect to deliver the final release with zero open defects.

6.3 Test Environment

All testing was performed on a Windows-based development machine running MongoDB locally. The three application layers were executed as real processes – the Vite development server on port 5173, the Express API on port 5000, and where relevant the FastAPI AI service on port 8000. The frontend was opened in Google Chrome, both as a desktop window

and in the browser's device-emulation mode so that mobile layouts could be checked without a physical phone.

Component	Environment Used
Processor / RAM	Intel Core i5-class laptop, 8 GB RAM (development baseline as per Chapter 3)
Operating system	Windows 10/11
Frontend runtime	React 19 + TypeScript 6, Vite dev server (port 5173), production build via "tsc -b && vite build"
Backend runtime	Node.js 20+ / Express 5 + TypeScript, executed with tsx (port 5000)
Database	MongoDB (local instance, mongodb://127.0.0.1:27017), Mongoose ODM
AI service	Python 3.12 + FastAPI (port 8000) – risk assessment end point
Browsers	Google Chrome (latest), including device-emulation for mobile widths
API testing tools	Command-line HTTP requests (curl) and in-browser network inspection
Static / build checks	tsc –noEmit (frontend + backend), oxlint, full Vite production build
Data verification	Direct inspection of MongoDB collections during and after each scenario

Keeping the database local and the services on the same machine made every test repeatable: the data created during a scenario could be inspected immediately in MongoDB, and any unexpected side effect could be traced to its source. Because some features – such as real SMS, email or WhatsApp delivery – depend on external providers that were not configured with credentials during testing, those paths were exercised as far as the application's honest status recording allows, meaning the system is verified to record the delivery status (for example NOT_CONFIGURED or FAILED) without ever corrupting the underlying incident record.

6.4 Test Strategy and Levels of Testing

Testing was carried out in layers, from the smallest unit of code up to the complete running system. Each level answers a different question, and a failure at a higher level could usually be traced back to a lower one.

6.4.1 Unit Level Verification

At the code level the project relies on TypeScript's strict compiler, which catches type errors, missing fields and incorrect usage across the entire codebase, together with the oxlint checker for code-quality problems. Data-sensitive utilities such as password hashing (bcrypt), JWT signing and verification, Indian phone-number normalization, distance calculation between coordinates, and the escaping of search text are exercised thoroughly through the API during integration testing. They are also the natural targets for the Jest/Supertest suite that the project intends to introduce as a permanent regression harness.

6.4.2 Integration Testing (API Level)

Every REST endpoint of the backend was tested against the running backend and MongoDB. This included the happy path, invalid inputs, missing authentication tokens, tokens of the wrong role, and requests for records that do not belong to the caller. After each operation the resulting documents were inspected directly in MongoDB to confirm that the database changed exactly as expected and nothing else. The role-based access-control matrix in Section 6.5.2 summarises these checks.

6.4.3 System Testing

The complete application – frontend, backend and database working together – was tested in a real browser. Every page was navigated, every form submitted, and every screen state (loading, empty, error, success) observed. This is the level where cross-cutting problems appear, such as an API returning data that the page could not render, or a redirect sending a user to the wrong place.

6.4.4 User Acceptance Testing

An end-to-end acceptance walkthrough of the complete user and administrator journey was executed in Google Chrome: sign-up, sign-in, emergency contact management, SOS incident creation, notification checking, the administrator dashboard, and role-based redirects. All nine steps of the walkthrough passed without any console error in the browser.

6.4.5 Regression Testing

After every defect fix the previously executed core scenarios were re-run to prove that the fix had not broken anything else. The most important example is the account access defect described in Section 6.8: after the fix, the complete acceptance walkthrough was executed again and passed end to end.

6.5 Types of Testing Performed

6.5.1 Functional Testing

Every functional requirement from Chapter 3 was converted into executable scenarios and verified in the running system. The results are documented module by module in Section 6.7. As part of functional verification the frontend was also compiled into a production build (TypeScript project build followed by a Vite production build) and the backend was verified to type-check cleanly, so the code that was functionally tested is the same code that would be deployed.

6.5.2 Security and Role-Based Access Control Testing

Security is the most important non-functional property of a women-safety system. The back end enforces three gates: authentication (a valid JWT must be presented), role authorization (administrator-only routes), and data ownership (a request may only touch the caller's own records). Each gate was tested with three kinds of request – no token, a normal-user token, and an administrator token – and the expected status codes were verified: 401 for missing or invalid authentication, 403 for a valid user attempting an administrator operation, and 200 for an authorized request. The matrix below summarises the access-control checks.

Endpoint (area)	No token	USER token	ADMIN token
POST /api/auth/login, GET /api/auth/me	401	200 (self)	200 (self)
/api/emergency-contacts (own contacts)	401	200 (own only)	200
/api/admin/dashboard	401	403	200
/api/admin/users	401	403	200
/api/admin/emergency-contacts (all contacts)	401	403	200
/api/admin/facilities, /api/admin/teams	401	403	200
/api/admin/incidents, assignments, reports	401	403	200
/api/admin/.../:id of another user's record	401	403	200 or 404

Beyond the matrix, ownership isolation was verified explicitly: a normal user received only their own emergency contacts, only their own incidents, and only their own notifications even when records belonging to other users existed in the database. Attempts to update or delete a record using another user's identifier were rejected. Passwords are stored only as bcrypt hashes, and login never reveals whether the failure was an unknown email or a wrong password.

6.5.3 Usability Testing

The interface was reviewed for clarity of labels, logical grouping of functions and recoverability from error states. Pages carry clear titles and descriptive text, forms mark required fields, and validation errors appear next to the offending field in the user's chosen language. The SOS flow is reachable in one tap from navigation on every user screen, and the placeholders and messages are available both in English and in Hindi through the language selector.

6.5.4 Compatibility and Responsiveness Testing

All main pages were opened at desktop width and then emulated at tablet (768 px) and mobile (approximately 375 px) widths. Layouts adapt as designed: the wide screen shows a tabular view of records, while narrower screens switch to stacked cards so that no information is cut off. The application is verified on Google Chrome at these widths; because the implementation is standards-based, it is expected to render consistently on other modern browsers.

6.5.5 Reliability and Data Integrity Testing

Several scenarios were executed to verify that the system keeps its records safe. A successfully submitted SOS incident was confirmed to persist immediately in MongoDB and to reappear in both the user's and the administrator's views. Creating an incident while location is unavailable did not fabricate a false location – the incident was saved with the location left genuinely absent. Financial-style integrity was also checked for the emergency-contact module: a user can mark exactly one primary contact, enforced by the database's partial unique index.

6.5.6 Performance Testing

Performance was observed on the local development machine during system testing. All core operations – login, loading the dashboard, listing contacts, submitting an SOS, and opening administrator pages – completed quickly, generally well below half a second on the local setup. Section 6.7 records these observations qualitatively per module; formal load and soak testing with many concurrent users was not part of the current exercise and is stated as a limitation in Section 6.10.

6.6 Test Data Preparation

All test data used in this chapter was genuine data created in the running system through the registration screen, the API, or dedicated seed scripts – nothing was hard-coded into the application to make screens look filled. Several dedicated accounts were registered covering

all three roles: normal users (including two users used specifically to prove that one user cannot see another's data), an administrator, and a responder. Test phone numbers used reserved Indian mobile-format numbers, and all email addresses were synthetic test addresses. Records (contacts, incidents, facilities, rescue teams, assignments, unsafe-area reports) were created, updated and deleted during testing, and the database was inspected after each scenario to confirm exactly what had been written. This approach means every test result below describes behaviour that was actually observed on real, running software.

6.7 Test Cases and Results (Module by Module)

The test cases below were executed against the running system. Each case gives a scripted description, the steps performed, the expected result, the result actually observed, and the final status. All cases listed were executed; a defect discovered in an earlier run (Section 6.8) was fixed and the corresponding cases were re-executed and passed. The abbreviation "ERR" in a step means the test expects the system to reject the input and show a clear message.

6.7.1 Authentication and Authorization

The authentication module is the gateway to the whole system. It was tested for correct registration, correct sign-in, token issuance, and the rejection of invalid attempts.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-01	User registration with valid details	Open /register; enter name, valid email, phone, password; submit	Account created; redirected to dashboard	PASS	PASS
TC-02	Duplicate email registration	Register again with the same email	Rejected with a clear validation message	PASS	PASS
TC-03	Login with correct credentials	Enter registered email and password; submit	JWT issued; redirected to role home	PASS	PASS
TC-04	Login with wrong password	Login with correct email but wrong password	401 "Invalid credentials."; no details leaked	PASS	PASS
TC-05	Login with unknown email	Login with an unregistered email	401 "Invalid credentials."	PASS	PASS
TC-06	Protected API without token	Request /api/auth/me with no Authorization header	401 Unauthorized	PASS	PASS
TC-07	USER on admin API	Use a normal-user JWT on /api/admin/dashboard	403 Forbidden	PASS	PASS
TC-08	ADMIN on admin API	Use an administrator JWT on /api/admin/dashboard	200 OK with dashboard data	PASS	PASS
TC-09	Logout	Click Logout from the header	Token cleared; protected pages redirect to /login	PASS	PASS

6.7.2 User and Profile Management

Profile management was verified for viewing and updating profile details and preferring a language.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-	View profile	Open the Profile page	Name, phone,	PASS	PASS

10			email and role shown correctly		
TC-11	Update display name and language	Change name and language (English/Hindi); save	Saved; interface labels switch language	PASS	PASS
TC-12	Empty required profile field	Attempt to save a blank name	ERR; validation message shown	PASS	PASS

6.7.3 Emergency Contacts

Emergency contacts are core to the SOS notification path. A user must manage only their own contacts, and alert settings must persist.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-13	Add an emergency contact	Open Contacts; enter name, phone, relation; enable SMS alert; save	Contact listed with badges	PASS	PASS
TC-14	Add contact with optional email	Repeat TC-13 with an email also entered	Email stored and shown	PASS	PASS
TC-15	Own contacts only	Compare two users' contact lists	Each user sees only their own contacts	PASS	PASS
TC-16	Edit a contact	Change phone and relation of a saved contact; save	Changes reflected; updatedAt changes	PASS	PASS
TC-17	Delete a contact	Delete an existing contact	Removed from list and from MongoDB	PASS	PASS
TC-18	Single primary contact	Mark two contacts as primary	Second attempt rejected; only one primary allowed	PASS	PASS
TC-19	Invalid phone number	Enter a non-Indian-format phone	ERR; clear validation message	PASS	PASS
TC-20	Admin sees all contacts	Open Admin – Emergency Contacts	All users' contacts listed with owner name/phone	PASS	PASS
TC-21	Admin edits/deletes a contact	Edit the phone; then delete the same contact	Change visible to the owner; deletion removes the record	PASS	PASS

6.7.4 SOS and Incident Management

The incident module is the heart of the system. It was verified for creation, persistence, history, status workflow and visibility rules.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-22	Create SOS incident	Open SOS; pick incident type/category; enter description; submit	Incident created with status REPORTED	PASS	PASS
TC-23	Incident persists immediately	Refresh the list after submission	Incident present; exists in MongoDB	PASS	PASS
TC-24	Incident saved without location	Create SOS while geolocation is denied	Created with no false location	PASS	PASS
TC-	Incident history	Change status through the	IncidentUpdate	PASS	PASS

25		workflow	records appended in order		
TC-26	Status workflow	Move REPORTED -> ACKNOWLEDGED -> ASSIGNED -> IN_PROGRESS -> RESOLVED -> CLOSED	Each transition accepted and recorded	PASS	PASS
TC-27	Cancel an incident	Cancel an open incident	Status becomes CANCELLED with a history entry	PASS	PASS
TC-28	User sees only own incidents	Open incident list as a normal user	Only own incidents shown	PASS	PASS
TC-29	Admin sees all incidents	Open Admin incidents with details	All incidents visible; filters work	PASS	PASS
TC-30	Empty/invalid description	Submit SOS with a blank description	ERR; validation message; nothing saved	PASS	PASS

6.7.5 Location Handling

Location is captured from the browser only with the user's permission, and its absence is handled honestly.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-31	Capture location on grant	Submit an incident while allowing geolocation	Latitude/longitude stored with the incident	PASS	PASS
TC-32	Location denied gracefully	Deny geolocation and submit	No coordinates stored; flow completes	PASS	PASS
TC-33	Location accuracy recorded	Inspect a saved location document	Accuracy value stored when provided	PASS	PASS

6.7.6 Facilities and Nearby Resources

Resource discovery was verified for the user and CRUD management for the administrator.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-34	List facilities	Open Resources; apply no filter	Hospitals/shelters/ others listed	PASS	PASS
TC-35	Search/filter resources	Type a keyword and filter by type	Matching results only	PASS	PASS
TC-36	Nearby resources	Request nearby facilities using a test location	Ordered list or graceful "unavailable" message	PASS	PASS
TC-37	Admin add/edit facility	Add a facility; edit its phone and status	Saved; visible to users	PASS	PASS
TC-38	Admin disable a facility	Set a facility to not operational	Marked in listing; not offered as active	PASS	PASS

6.7.7 Rescue Teams and Assignments

Team records and software-record assignments were verified, with administrator-only access.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-39	Admin register/add a team	Add a rescue team with type and phone	Team listed in Admin Teams	PASS	PASS
TC-40	USER cannot manage teams	Attempt team management with a USER JWT	403 Forbidden	PASS	PASS
TC-41	Create an assignment	Assign a team to an open incident	Assignment recorded with status ASSIGNED	PASS	PASS
TC-42	Assignment lifecycle	Move ASSIGNED -> EN_ROUTE -> ON_SCENE -> COMPLETED	Each transition recorded	PASS	PASS
TC-43	Invalid incident/team id	Assign using a malformed ObjectId	400 validation error	PASS	PASS

6.7.8 Notifications

Notification behaviour was tested both with real in-app delivery and with external channels honestly reported when unconfigured.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-44	In-app notification on incident event	Create an incident; open Notifications	New in-app notification appears	PASS	PASS
TC-45	Unread count	Refresh after an unread notification	Bell badge shows the unread count	PASS	PASS
TC-46	Contact notified via SMS if configured	Create incident with SMS-enabled primary contact	Notification recorded; SENT/DELIVERED or FAILED per provider	PASS	PASS
TC-47	Unconfigured channel handled honestly	Trigger a channel with no provider set	Status NOT_CONFIGURED/UNAVAILABLE; incident unaffected	PASS	PASS

6.7.9 Unsafe-Area Reporting

Reports are moderated: submission is open, but confirmation happens only after review.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-48	Submit unsafe-area report	Submit a report with location, category, description	Record created; marked not-verified	PASS	PASS
TC-49	Admin review and verify	Open Admin Unsafe Reports; verify the report	isVerified becomes true; visible as verified	PASS	PASS
TC-50	Not auto-confirmed	Check a fresh report without review	Not treated as confirmed until verified	PASS	PASS

6.7.10 AI Risk Assessment

The risk assessment returns decision-support scores whose structure and range were verified.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-51	Risk score in range	Trigger assessment for a test location	Score 0–100 returned	PASS	PASS
TC-52	Risk level mapping	Check the level returned with the score	LOW/MEDIUM/HIGH/CRITICAL mapped consistently	PASS	PASS
TC-53	Service unavailable gracefully	Stop the AI service; request an assessment	Clear message; core app unaffected	PASS	PASS

6.7.11 Admin Functions – Dashboard, Users, Reports, Logs

Administrator functions were verified for correctness against real stored data and strict role gating.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-54	Dashboard statistics	Open Admin Dashboard	Counts match the actual MongoDB data	PASS	PASS
TC-55	User listing with search	Search users by name/phone; page through results	Correct filtered paginated results	PASS	PASS
TC-56	User detail with incidents/contacts	Open a user's detail page	Only that user's related records shown	PASS	PASS
TC-57	Analytics/reports view	Open Admin Reports	Statistics derived from stored data	PASS	PASS
TC-58	Report export	Generate a report and export	PDF/CSV/JSON produced from data snapshot	PASS	PASS
TC-59	Admin activity logging	Perform several admin actions	AdminLogs entries created with action/target/time	PASS	PASS
TC-60	Logs restricted to admins	Request admin logs with a USER token	403 Forbidden	PASS	PASS
TC-61	Admin Emergency Contacts page	Open Admin Emergency Contacts; search	All contacts with owners listed; search works	PASS	PASS

6.7.12 Validation, Edge Cases and Data Integrity

A final round of cases attacked the system with malformed and unusual input.

TC ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC-62	Empty required fields	Submit each main form with fields left blank	ERR on every required field; nothing persisted	PASS	PASS
TC-63	Whitespace-only input	Submit fields containing only spaces	Treated as empty and rejected	PASS	PASS

TC-64	Special characters in search	Search with characters such as <code>()[]*+</code>	Returned results or empty list; no error/crash	PASS	PASS
TC-65	Very long input	Enter text well beyond field limits	Rejected with a length-validation message	PASS	PASS
TC-66	Malformed ObjectId	Use an invalid id in a path parameter	400 validation error, not a crash	PASS	PASS
TC-67	Non-existent record id	Update/delete a valid-shaped id that does not exist	404 not found	PASS	PASS
TC-68	Phone normalization	Enter 10-digit, +91 and spaces variants	Normalized to the same valid form	PASS	PASS
TC-69	Unicode / Hindi input	Save a name written in Hindi	Stored and displayed correctly	PASS	PASS
TC-70	Ownership isolation (cross-user)	Attempt to read/delete another user's contact via its id	Rejected; target data not exposed	PASS	PASS

6.8 Defects Found and Fixed During Testing

Testing is valuable only when it actually finds problems. Two genuine defects were found during this exercise. Both were fixed, and the affected scenarios were re-executed successfully. They are documented below with their root cause and the verification of the fix, which is the pattern followed whenever a defect is reported.

Defect 1 – Valid administrator account appeared “deactivated”

Symptom: A legitimate administrator whose account was created directly in the database could not log in (the API returned “Invalid credentials.”) and every administrator API request returned “Account is deactivated.” The account’s credentials were correct, yet the application was sending the user back to the login page every time.

Root cause: The account document had been inserted without the `isActive` field. The login and authorization logic checked “is the account deactivated?” using a strict comparison that treated a missing value as deactivated. Because the boolean field was undefined on this record, a perfectly active administrator was classified as disabled.

Fix: The deactivation checks in the login controller and in the authorization middleware were corrected to treat a record as deactivated only when the field is explicitly false. This preserves the behaviour for accounts that are genuinely switched off while restoring access for active records that simply predate the field.

Verification: The administrator was then able to authenticate, and the previously failing endpoints (`/api/auth/me`, `/api/admin/dashboard`, `/api/admin/users`) all returned 200. The full browser acceptance walkthrough was re-run and passed all nine steps, confirming no regression.

Defect 2 – Emergency-contacts administration page routed to login

Symptom: Navigating to the administrator’s Emergency Contacts destination fell back to the login page because the client-side hash router did not recognise the route, so no page component was rendered.

Root cause: The page, its route entry in the hash router and the navigation menu item had not been wired into the front end, even though the backend endpoints supporting the page already existed.

Fix: The admin Emergency Contacts page was implemented, the route was added to the typed route list and the known-routes table, and a navigation entry was added to the administrator menu. The page fetches all emergency contacts with the owner's name, phone and email, supports search and pagination, offers a call (tel:) action, and allows the administrator to edit or delete any contact through the protected admin API.

Verification: With an administrator token the endpoint returned 200 with all contacts; with a normal-user token it returned 403, and without a token 401. Editing a contact from the admin page was confirmed to reflect for the owning user, and deleting it removed the real record from MongoDB.

Separately, minor hardening was added during testing: search text is escaped before building regular expressions, so special characters cannot break queries or be abused as injection patterns. This was verified with a dedicated case (TC-64).

6.9 Test Summary

A total of seventy test cases were designed across the twelve areas described above, covering the full functional scope of the system together with security, validation and edge-case behaviour. Every case was executed against the running application. The summary below shows the distribution and the overall pass rate.

Test Area	Cases Executed	Passed	Failed
Authentication and authorization	9	9	0
User and profile management	3	3	0
Emergency contacts	9	9	0
SOS and incident management	9	9	0
Location handling	3	3	0
Facilities and nearby resources	5	5	0
Rescue teams and assignments	5	5	0
Notifications	4	4	0
Unsafe-area reporting	3	3	0
AI risk assessment	3	3	0
Admin functions (dashboard/users/reports/logs)	8	8	0
Validation, edge cases and data integrity	9	9	0
Total	70	70	0

Overall pass rate: 100% (70 of 70 cases). The two defects discovered during the exercise were resolved and their areas re-tested, which is why the final totals show no open failures. These results cover the functional, security, usability, compatibility, reliability and basic performance dimensions of the system on the documented test environment.

6.10 Limitations of the Testing Exercise

It is important to state what this testing exercise did not cover, so that the confidence it provides is not overstated:

All testing ran on a single local development machine with MongoDB installed locally. No multi-machine, staging or production environment was exercised.

External communication providers (real SMS, real email, WhatsApp) were not configured with live credentials. Their failure paths were verified through the application's status-recorded behaviour (NOT_CONFIGURED / FAILED / UNAVAILABLE), but real end-to-end message delivery was not measured.

Performance was observed on local hardware with light concurrent use; formal load testing, soak testing and benchmarking at scale were not performed.

The AI risk assessment was verified for service availability, output structure and score range; its analytical accuracy is decision-support software and was not validated against ground truth.

Browser coverage was concentrated on Google Chrome at desktop, tablet and mobile emulation widths; other browsers were not separately exercised.

A permanent automated regression suite (Jest/Supertest, browser test runner) is not yet committed to the repository. The repeatable checks currently used – typechecking, linting, production builds and the scripted API and browser walkthroughs – are the foundation that such a suite will codify.

6.11 Chapter Summary and Conformance

This chapter validated **Raksha Safe** across code, API, and full-browser levels against specified requirements. Key outcomes include:

- **100% Test Success:** All 70 test cases passed cleanly on the documented environment, with every identified defect resolved and re-verified.
- **Robust Security:** Authentication, role-based access control (RBAC), and user data isolation were successfully enforced (verified via 401/403/200 checks).
- **Technical Stability:** Frontend and backend TypeScript builds passed without errors, functional modules operated smoothly with real MongoDB data, and validation reliably handled malformed inputs.
- **Flawless E2E Execution:** End-to-end user and administrator workflows completed in a real browser with zero console errors or crashes.